

INDR-100 HOMEWORK 2
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Question-1

1. (20 pts) An investor has a limited budget and wants to invest in various projects. However, each project not only requires capital investment but also requires a certain number of employees. The investor has both budget and workforce constraints.

This means each potential investment opportunity (project j) is characterized by three key attributes: an expected profit ($profit_j$, measured in thousand TL), the capital investment required ($capital_j$, in thousand TL), and the number of employees from the investor's team needed to support that project ($employees_j$, number of people).

The investor faces two critical constraints. First, there is a limited budget of B thousand TL available for investments. Second, the investor's company has only E employees available to be assigned across all selected projects.

An employee assigned to one project cannot work on another, so the total number of employees needed across all chosen projects cannot exceed the available workforce.

The investor must decide which projects to undertake to maximize total profit while staying within both budget and workforce constraints.

- Write the complete mathematical formulation including: (i) Defining appropriate decision variables, (ii) Objective function, (iii) Budget constraint, (iv) Workforce constraint, and (v) Variable type constraints. Explain each of them verbally.
- Suppose there are 5 projects with the following data. Write the specific mathematical model for this problem instance using the data below.

Project	Profit (1000 TL)	Capital (1000 TL)	Employees
1	50	30	5
2	40	20	8
3	60	35	6
4	30	15	4
5	45	25	7

Budget: $B = 60$ thousand TL, Workforce: $E = 12$ employees

Answers:

a) Mathematical Model:

Decision Variables:

$$x_j = \begin{cases} 1, & \text{if project } j \text{ is selected} \\ 0, & \text{else} \end{cases}, \quad j = 1, 2, \dots, n$$

(x_j is a binary decision variable ; decision variables j)

Variable Type

$$x_i \in \{0, 1\}, \forall j$$

(Variables can only take the value 0 or 1)

Objective Function:

$$\max Z = \sum_{j=1}^n (profit_j x_j)$$

(We need to maximize total profit)

Budget Constraint:

$$\sum_{j=1}^n (capital_j x_j) \leq (B)udget \text{ Constraint}$$

(The total budget of the selected project can not be more than the available budget)

Employee Constraint:

$$\sum_{j=1}^n (Employees_j x_j) \leq (E)mployees \text{ Constraint}$$

(The total number of employees assigned the selected project can not be higher than the available budget)

b) Linear Programming Model:

	Project	Profit (1000€)	CAPITAL (1000€)	Employees
$x_1 \rightarrow$	1	50	30	5
$x_2 \rightarrow$	2	40	20	8
$x_3 \rightarrow$	3	60	35	6
$x_4 \rightarrow$	4	30	15	4
$x_5 \rightarrow$	5	45	25	7

Maximize: $Z = 50x_1 + 40x_2 + 60x_3 + 30x_4 + 45x_5$

Subject To :

$$30x_1 + 20x_2 + 35x_3 + 15x_4 + 25x_5 \leq 60 = \text{Budget Constraint}$$

$$5x_1 + 8x_2 + 6x_3 + 4x_4 + 7x_5 \leq 12 = \text{Employee Constraint}$$

$$x_1, x_2, x_3, x_4, x_5 \in \{0,1\}$$

The decision variables show the number of employers and these are determined by constraints, it will effect the objection value , in addition this model shows us optimal solution for the problem since equation is exteed to constraints (Budget and Workforce).

Question-2

2. (35 pts) You are given the following linear programming model:

$$\begin{aligned} \max \quad & Z = -100x_1 + 50x_2 \\ \text{st:} \quad & 6x_1 + 6x_2 \leq 72 & (C1) \\ & -6x_1 + 3x_2 \leq 27 & (C2) \\ & 10x_1 \leq 100 & (C3) \\ & x_1 - 2x_2 \leq 8 & (C4) \\ & x_1 \geq 0 & (C5) \\ & x_2 \geq 0 & (C6) \end{aligned}$$

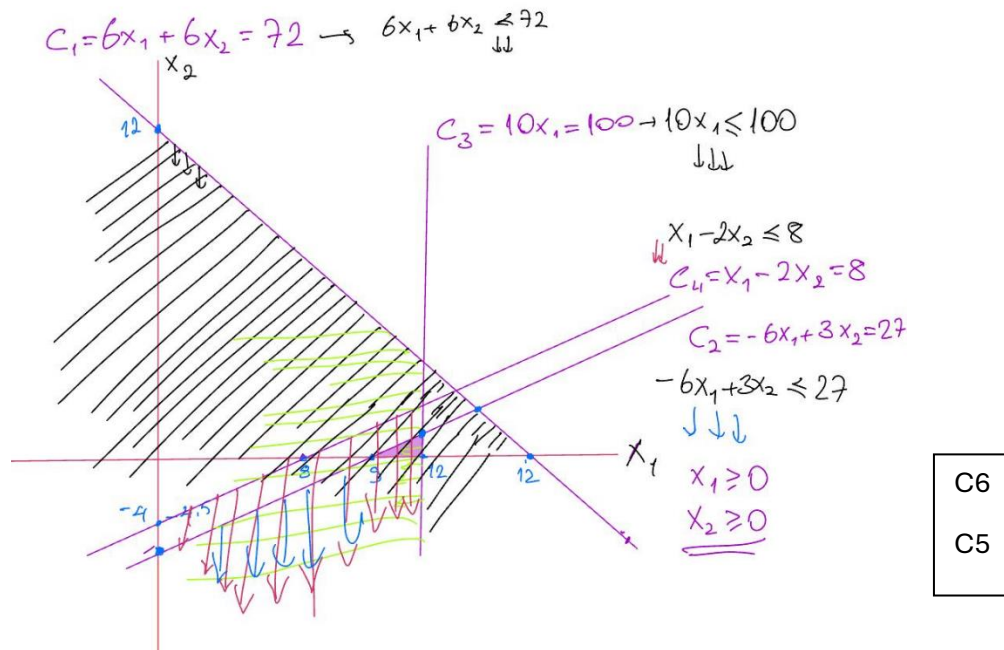
- Draw the graphical representation of the model indicating each constraint. Identify the feasible region on the graph. *Note: Make sure that you clearly label all the constraints and feasible region on the graph.*
- Give an upper bound on the number of corner point solutions of the problem by using the formula.
- Find all candidate corner points by intersecting each pair of constraint boundary lines, including the non-negativity constraints. Complete the table below by writing the intersecting constraints, the resulting point, the feasibility check, and the final status as feasible, infeasible, or inconsistent. Show the feasible corner points on the graph and label them clearly. *Note: In the "Feasibility Check" column, state whether the point satisfies all constraints or specify the violated constraint(s), such as "all constraints satisfied" or "violates C2"*

#	Intersection of	Corner Point	Feasibility Check	Status
1				
2				
...				

- Find the optimal solution(s) to the given model using the corner point method (show your equations and calculations).
- Draw the objective function contour on the graph where it is equal to optimal objective value.

Answers:

a)



b) Upper Bound Formula:

a = number of constraints = 4

b = number of variables = 2

We may include or exclude non-negativity constraints but we have to state this version.

$$\text{Upper Bound} = \binom{m+n}{n} = \binom{6}{2} = 15$$

There are 6 lines (constraints). We need 2 lines to create one corner point, so at most 15 points can be created.

c) All candidate corner points

1- All intersection points:

C1-C2(1,11) C2-C3(10,29) C3-C4(10,1) C4-C5(0,-4) C5-C6(0,0)

C1-C3(10,2) C2-C4(-8.67,-8.33) C3-C5(Parallel) C4-C6(8,0)

C1-C4(10.6,1.3) C2-C5(0,9) C3-C6(10,0)

C1-C5(0,12) C2-C6(-4.5,0)

C1-C6(12,0)

(C2-C6),(C2-C3),(C3-C6) They are feasible(Corner) points

d) Optimal Solution-Corner Point

$$Z = -100x_1 + 50x_2$$

$$C2-C6 = -100(0) + 50(9) = 450$$

$$C1-C3 = -100(10) + 50(2) = -900$$

$$C3-C6 = -100(10) + 50(0) = -1000$$

Optimal Point is C2-C6(0,9). This means that the maximum value of Z is obtained at the (0,9) point.

e) I draw the answer in the graph of (a)

$$Z = 450$$

Question-3

3. (25 pts) A university academic advisor is studying the relationship between weekly study hours and exam performance to provide better guidance to students. The advisor collected data from five students in an introductory engineering course and recorded their average weekly study hours and their final exam scores (out of 100 points):

Student	Weekly Study Hours (x)	Exam Score (y)
1	5	55
2	8	70
3	10	78
4	12	86
5	15	95
6	15	0

The advisor wants to develop a linear regression model.

- Using the data from these five students, manually calculate the linear regression model of the form $y = a + bx$, where y represents the exam score and x represents weekly study hours. Show all your calculations step by step. Specifically, determine: (i) slope, (ii) intercept, and (iii) complete regression equation.
- Calculate the coefficient of determination (R^2). Comment on how well this model explains the data based on this coefficient.
- Now, show if there is any outlier in your model (you can use two standard deviation rule). If so, remove the outlier from the data and recalculate the model parameters and the coefficient of determination (R^2). Comment on the results.

Answers:

a) Regression :

Student	x	y	x^2	xy
1	5	55	25	275
2	8	70	64	560
3	10	78	100	780
4	12	86	144	1032
5	15	95	225	1425
Σ	50	384	558	4072

Slope

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b = \frac{5(4072) - 50(384)}{5(558) - (50)(50)} = \frac{1160}{290} = 3,97$$

Intercept

$$a = \bar{y} - bx$$

$$a = \frac{384}{5} - 4\frac{50}{5} = 36.8$$

$$\text{regression equation: } \hat{y} = 36.8 + 4x$$

When students work an extra hour, their exam score will increase 4 point added to the average

b) R^2

$$SS_{res} = \sum (y_i - \hat{y}_i)^2$$

$$SS_{tot} = \sum (y_i - \bar{y})^2$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

For each one of student there is different $\hat{y}$

c) Outlier detection

there are $x=15$ students

Question-4

4. (20 pts) A popular food truck parks outside a university campus every weekday at noon. The truck has a single employee who takes and prepares each order individually, from start to finish. Orders are served in first-come, first-served order. No customers are present when the truck begins setup and no customer leaves the queue once they join. Although the truck arrives and sets up at 12:00 PM (time 0), the employee only begins serving customers at 12:05 PM (time 5) due to the setup routine. Customers who arrive before the truck starts serving will wait in line until service begins.

The interarrival times and service times (in minutes) for the first five customers of the day are recorded below:

Customer	Interarrival time (min)	Service time (min)
1	3	7
2	3	4
3	3	3
4	7	5
5	5	2

- a) For each of the five customers, determine: (i) arrival time, (ii) service start time, (iii) service completion (departure) time, and (iv) waiting time in queue.
- b) Using the results from part a, compute: (i) the average waiting time across all customers, (ii) the maximum number of customers waiting in queue at any point in time, and (iii) the server utilization during the period from the first arrival until the last departure.

Note: Just results placed in the table will not be graded. You should to show how you calculate each.

Answers:

a) Arrival time = previous arrival time + interarrival time

service start at 12.05 with $t=5$ all time are in this table:

Customer	Arrival	Service Start	Service End	Waiting Time
1	3	5	$5 + 7 = 12$	$5 - 3 = 2$
2	$3 + 3 = 6$	$\max(12, 6) = 12$	$12 + 4 = 16$	$12 - 6 = 6$
3	$6 + 3 = 9$	$\max(16, 9) = 16$	$16 + 3 = 19$	$16 - 9 = 7$
4	$9 + 7 = 16$	$\max(19, 16) = 19$	$19 + 5 = 24$	$19 - 16 = 3$
5	$16 + 5 = 21$	$\max(24, 21) = 24$	$24 + 2 = 26$	$24 - 21 = 3$

Service start = $\max$ (previous customer's service end time, this customer's arrival time:

1-12

2-16

3-19

4-24

5-26

b)Statistics:

Average waiting time = $\bar{W} = \frac{2+6+7+3+3}{5} = 4.2$

Utilization:

$$\text{Utilization} = \frac{\text{Total busy time}}{\text{Total time}}$$

$$\text{Utilization} = \frac{7 + 4 + 3 + 5 + 2}{26 - 3}$$

$$\text{Utilization} = \frac{21}{23} \approx 0.913 = 91.3\%$$

